AZIM PREMJI UNIVERSITY, BHOPAL

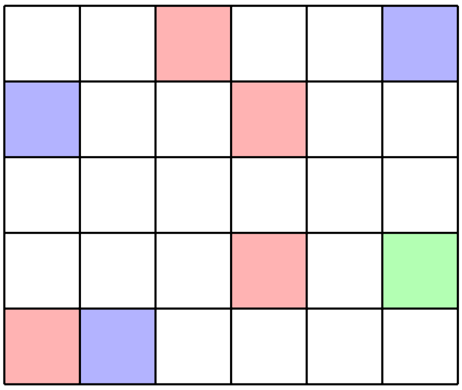
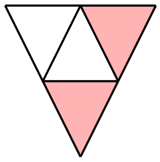
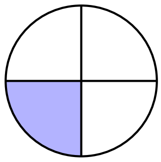
First Semester 2026-2027

Tutorial Sheet-2

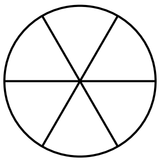
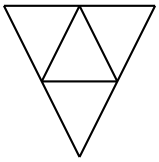
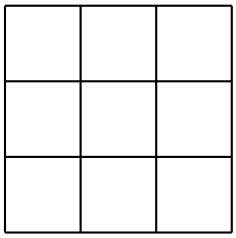
Course No. MTH-113 Course title: Introduction to Quantitative Thinking

---

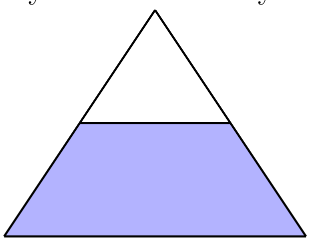
(1) Write the fraction representing the shaded reason of each color in the below pictures

(a)  (b)  (c) 

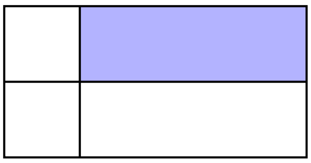
(2) Shade the part according to the given fraction.

(a) $\frac{1}{6}$  (b) $\frac{1}{4}$  (c) $\frac{4}{9}$ 

(3) Identify the errors if any



(a) The shaded region is $\frac{1}{2}$



(b) The shaded region is $\frac{1}{4}$

(4) (a) What fraction of a day is 8 hours?
(b) What fraction of an hour is 40 minutes?

(5) Arya, Abhimanyu, and Vivek shared lunch. Arya has brought two sandwiches, one made of vegetable and one of jam. The other two boys forgot to bring their lunch. Arya agreed to share his sandwiches so that each person will have an equal share of each sandwich.
(a) How can Arya divide his sandwiches so that each person has an equal share?
(b) What part of a sandwich will each boy receive?

(6) Kanchan dyes dresses. She had to dye 30 dresses. She has so far finished 20 dresses. What fraction of dresses has she finished?

(7) Write the natural numbers from 2 to 12. What fraction of them are prime numbers?

(8) A fraction is given.
How will you decide, by just looking at it, whether, the fraction is

(a) less than 1?
(b) equal to 1?

(9) Fill up using one of these: “>”, “<” or “=”

(a) $\frac{1}{2} \square 1$ (b) $\frac{3}{5} \square 1$ (c) $1 \square \frac{7}{8}$ (d) $\frac{4}{4} \square 1$

(10) Draw number lines and locate the points on them:

(a) $\frac{1}{2}, \frac{1}{4}, \frac{3}{4}, \frac{4}{4}$ (b) $\frac{1}{8}, \frac{2}{8}, \frac{3}{8}, \frac{7}{8}$ (c) $\frac{2}{5}, \frac{3}{5}, \frac{8}{5}, \frac{4}{5}$

(11) Express the following as mixed fractions:

(a) $\frac{20}{3}$ (b) $\frac{11}{5}$ (c) $\frac{17}{7}$ (d) $\frac{28}{5}$ (e) $\frac{19}{6}$ (f) $\frac{35}{9}$

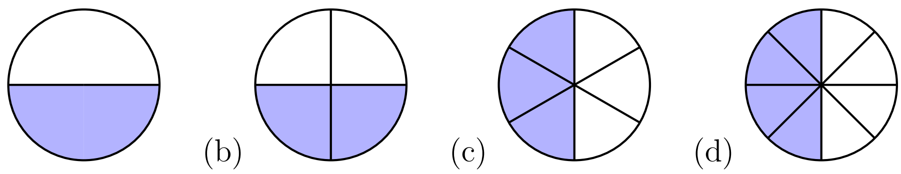
(12) Express the following as improper fractions:

(a) $7\frac{3}{4}$ (b) $5\frac{6}{7}$ (c) $2\frac{5}{6}$ (d) $10\frac{3}{5}$ (e) $9\frac{3}{7}$ (f) $8\frac{4}{9}$

(13) Find five equivalent fractions of each of the following:

(a) $\frac{2}{3}$ (b) $\frac{1}{5}$ (c) $\frac{3}{5}$ (d) $\frac{5}{9}$

(14) Write the fractions. Are all these fractions equivalent?



(a) (b) (c) (d)

(15) Reduce the following fractions to simplest form:

a) $\frac{48}{60}$ (b) $\frac{150}{60}$ (c) $\frac{84}{98}$ (d) $\frac{12}{52}$ (e) $\frac{7}{28}$

(16) Compare the fractions and put an appropriate sign.

(a) $\frac{3}{6} \square \frac{5}{6}$ (b) $\frac{1}{7} \square \frac{1}{4}$ (c) $\frac{4}{5} \square \frac{5}{5}$ (d) $\frac{3}{5} \square \frac{2}{3}$ (e) $\frac{3}{4} \square \frac{2}{8}$ (f) $\frac{3}{7} \square \frac{4}{9}$

(17) Ila read 25 pages of a book containing 100 pages. Lalita read 25 of the same book. Who read less?

(18) In a class A of 25 students, 20 passed in first class; in another class B of 30 students, 24 passed in first class. In which class was a greater fraction of students getting first class?

(19) Fill in the missing fractions.

(a) $\frac{7}{10} - \square = \frac{3}{10}$ (b) $\square - \frac{3}{21} = \frac{5}{21}$ (c) $\square - \frac{3}{5} = \frac{3}{5}$ (d) $\square + \frac{5}{27} = \frac{12}{27}$

(20) (a) Javed was given $\frac{5}{7}$ of a basket of oranges. What fraction of oranges was left in the basket?
(b) Solve
(i) $\frac{2}{3} + \frac{1}{7}$ (ii) $\frac{3}{10} + \frac{7}{15}$ (iii) $\frac{4}{9} + \frac{2}{7}$ (iv) $\frac{2}{3} + \frac{3}{4} + \frac{1}{2}$ (v) $4\frac{2}{3} + 3\frac{1}{4}$ (vi) $\frac{1}{2} + \frac{1}{3} + \frac{1}{6}$

****